In order to meaningfully compare the local space-expansion rates at two distant points in the universe, you first have to know exactly *how* distant each one is. This is no big deal for something on Earth, or even something as close as the Moon, since you can measure the distance by bouncing a laser beam off of it and seeing how long the light takes to come back.[fn6] On those kinds of scales, the universe is pretty reasonable. It acts basically like an unchanging space where the distance from A to B is straightforwardly measurable and makes sense and everything works. When it comes to things outside the Solar System, it gets trickier, both because things that are more distant are harder to measure, and because on larger and larger scales, the expansion starts to change the definition of distance itself.

Astronomers have, over the years, patched together with duct tape and twine a set of overlapping definitions and measurements of distance that build upon one another. As kludgy as it still sometimes seems, it's the result of decades of innovations in observational astronomy and data analysis, and has given us an intuitive but frustratingly difficult-to-implement strategy known as the *distance ladder*.

Let's say you need to measure the length of a large room, and all you have is an ordinary-sized ruler. You could lay the ruler down repeatedly until you cover the length of the room, if you don't mind crawling around on the floor. Or you could be a bit more creative and measure the length of your stride, then just walk across the room, counting steps. If you chose the steps method, you're creating a distance ladder: a system of measuring a large distance by calibrating your measurements with something more manageable.

In astronomy, the distance ladder has a series of rungs that allow it to extend out to objects that are billions of light-years away. Within the Solar System, direct laser measurements, orbital scalings, and even eclipses help us gather distance data. Beyond that, the next step is to use parallax. This is a method that takes into account the fact that when you change your vantage point, nearby things seem to shift their positions relative to a fixed background more than distant ones do. It's the same effect that makes a finger held in front of your face seem to jump back and forth when you close one eye and then the other. If we look at a nearby star in June, and then the same star in December, the fact that the Earth is in a different location in its orbit around the Sun means that the star will appear to have